B. Crimson Triples

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 900 | Tags: dp, math, number theory | https://codeforces.com/problemset/problem/2238/B

Chills run down your spine...

— Terraria

After summoning the next boss — The Brain of Cthulhu, you noticed that it surrounds itself with n eyes, numbered from 1 to n. In one attack, The Brain of Cthulhu chooses a triple of eyes (not necessarily distinct) with numbers (a, b, c). The triple of eyes is called *crimson* if and only if the following property holds:

$\gcd^*(\mathrm{lcm}^\dagger(a, b), \mathrm{lcm}(b, c)) = \gcd(a, c)$,

To defeat the boss, you want to know how many ways The Brain of Cthulhu can choose a crimson triple of eyes. The triples of eyes (a_1, b_1, c_1) and (a_2, b_2, c_2) are considered different if $a_1 \neq a_2$, or $b_1 \neq b_2$, or $c_1 \neq c_2$.

$^*\gcd(x, y)$ denotes the greatest common divisor (GCD) of integers x and y.

$^\dagger\mathrm{lcm}(x, y)$ denotes the least common multiple (LCM) of integers x and y.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 1000$). The description of the test cases follows.

The only line of each test case contains one integer n ($1 \le n \le 2 \cdot 10^5$) — the number of eyes of The Brain of Cthulhu.

It is guaranteed that the sum of n across all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output one integer — the number of ways to choose a crimson triple of eyes.

Example

Input
3 1 2 20

Output
1 5 612

Note

In the first test case, there is 1 possible crimson triple: (1, 1, 1).

In the second test case, there are 5 possible crimson triples: (1, 1, 1), (1, 1, 2), (2, 1, 1), (2, 1, 2), (2, 2, 2).